HUNDRED DAYS OF CODE
DAY 9

QUESTION 1:

Write a **menu-driven** C program to input two numbers and swap them using a third variable & without using third variable. Also display the values before and after swapping.

Algorithm

1. Start.
2. Display the menu:
   - Swap using a third variable
   - Swap without a third variable
   - Exit
3. Read the user's choice.
4. If swapping is selected, input two numbers a and b.
5. Display the values before swapping.
6. If choice is 1:
   - Store a in temp.
   - Assign b to a.
   - Assign temp to b.
7. If choice is 2:
   - a = a + b
   - b = a - b
   - a = a - b
8. Display the values after swapping.
9. Repeat the menu until Exit is selected.
10. Stop.

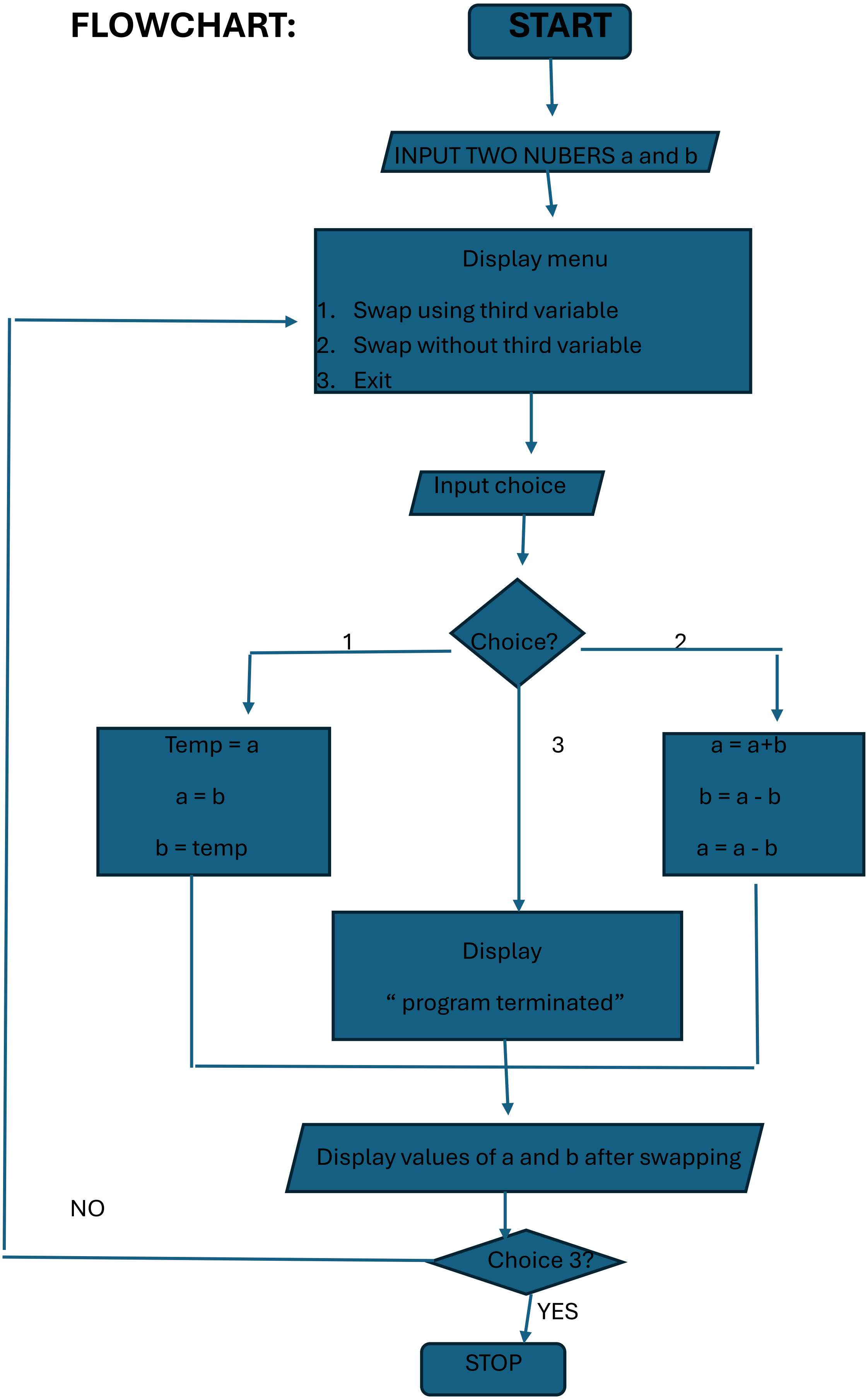
HUNDRED DAYS OF CODE
DAY 9

Pseudocode

```
REPEAT
    DISPLAY "1. Swap using third variable"
    DISPLAY "2. Swap without third variable"
    DISPLAY "3. Exit"
    INPUT choice

    IF choice = 1 OR choice = 2 THEN
        INPUT a, b
        DISPLAY a, b before swapping

        IF choice = 1 THEN
            temp ← a
            a ← b
            b ← temp
        ELSE
            a ← a + b
            b ← a - b
            a ← a - b
        END IF
        DISPLAY a, b after swapping

    ELSE IF choice ≠ 3 THEN
        DISPLAY "Invalid choice"
    END IF
UNTIL choice = 3
END
```

HUNDRED DAYS OF CODE
DAY 9

FLOWCHART:



HUNDRED DAYS OF CODE
DAY 9

REQUIRED C PROGRAM:

```
#include <stdio.h>


int main()
{
    int a, b, temp, choice;


    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);


    do
    {
        printf("\n--- SWAPPING MENU ---\n");
        printf("1. Swap using third variable\n");
        printf("2. Swap without using third variable\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);


        switch(choice)
        {
            case 1:
                printf("\nBefore swapping: a = %d, b = %d\n", a, b);


                temp = a;
                a = b;
                b = temp;
```

HUNDRED DAYS OF CODE
DAY 9

```
            printf("After swapping:  a = %d, b = %d\n", a, b);
            break;

        case 2:
            printf("\nBefore swapping: a = %d, b = %d\n", a, b);

            a = a + b;
            b = a - b;
            a = a - b;

            printf("After swapping:  a = %d, b = %d\n", a, b);
            break;

        case 3:
            printf("Program terminated.\n");
            break;

        default:
            printf("Invalid choice! Please try again.\n");
        }

    } while(choice != 3);

    return 0;
}
```

HUNDRED DAYS OF CODE
DAY 9

TEST CASE TABLE:

Test Case	**a**	**b**	**Choice**	**Expected Output**
TC-01	10	20	1	a = 20, b = 10
TC-02	25	15	2	a = 25, b =15
TC-03	-8	12	1	a = -8, b = 12

COMPILATION AND EXECUTION

```
┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ ./swapping
═════ MENU ═════
1. Swap using Third Variable
2. Swap without Third Variable
Enter your choice: 1
Enter First Number: 10
Enter Second Number: 20

Before Swapping:
First Number = 10
Second Number = 20

After Swapping:
First Number = 20
Second Number = 10
```

```
┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ ./swapping
═════ MENU ═════
1. Swap using Third Variable
2. Swap without Third Variable
Enter your choice: 2
Enter First Number: 25
Enter Second Number: 15

Before Swapping:
First Number = 25
Second Number = 15

After Swapping:
First Number = 15
Second Number = 25
```

HUNDRED DAYS OF CODE
DAY 9

```
┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ ./swapping
═════ MENU ═════
1. Swap using Third Variable
2. Swap without Third Variable
Enter your choice: 1
Enter First Number: -8
Enter Second Number: 12

Before Swapping:
First Number = -8
Second Number = 12

After Swapping:
First Number = 12
Second Number = -8
```

```
┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ vi swapping.c

┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ gcc swapping.c -o swapping

┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ ./swapping
═════ MENU ═════
1. Swap using Third Variable
2. Swap without Third Variable
Enter your choice: 1
Enter First Number: 10
Enter Second Number: 20

Before Swapping:
First Number = 10
Second Number = 20

After Swapping:
First Number = 20
Second Number = 10

┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ ./swapping
═════ MENU ═════
1. Swap using Third Variable
2. Swap without Third Variable
Enter your choice: 2
Enter First Number: 25
Enter Second Number: 15

Before Swapping:
First Number = 25
Second Number = 15
```

```
After Swapping:
First Number = 15
Second Number = 25

┌──(kali㉿kali)-[~/Desktop/exp8
└─$ ./swapping
═════ MENU ═════
1. Swap using Third Variable
2. Swap without Third Variable
Enter your choice: 1
Enter First Number: -8
Enter Second Number: 12

Before Swapping:
First Number = -8
Second Number = 12

After Swapping:
First Number = 12
Second Number = -8
```

HUNDRED DAYS OF CODE
DAY 9

QUESTION 2 - Write a **menu-driven** C program to find the sum of the first n natural numbers and sum of squares of the first n natural numbers.

Algorithm

1. Start.
2. Input the value of n.
3. Display the menu.
4. Read the user's choice.
5. If choice is 1, calculate the sum of first n natural numbers.
6. If choice is 2, calculate the sum of squares of first n natural numbers.
7. Display the calculated result.
8. If choice is 3, exit.
9. Display an error for an invalid choice.
10. Stop.

PSEUDOCODE

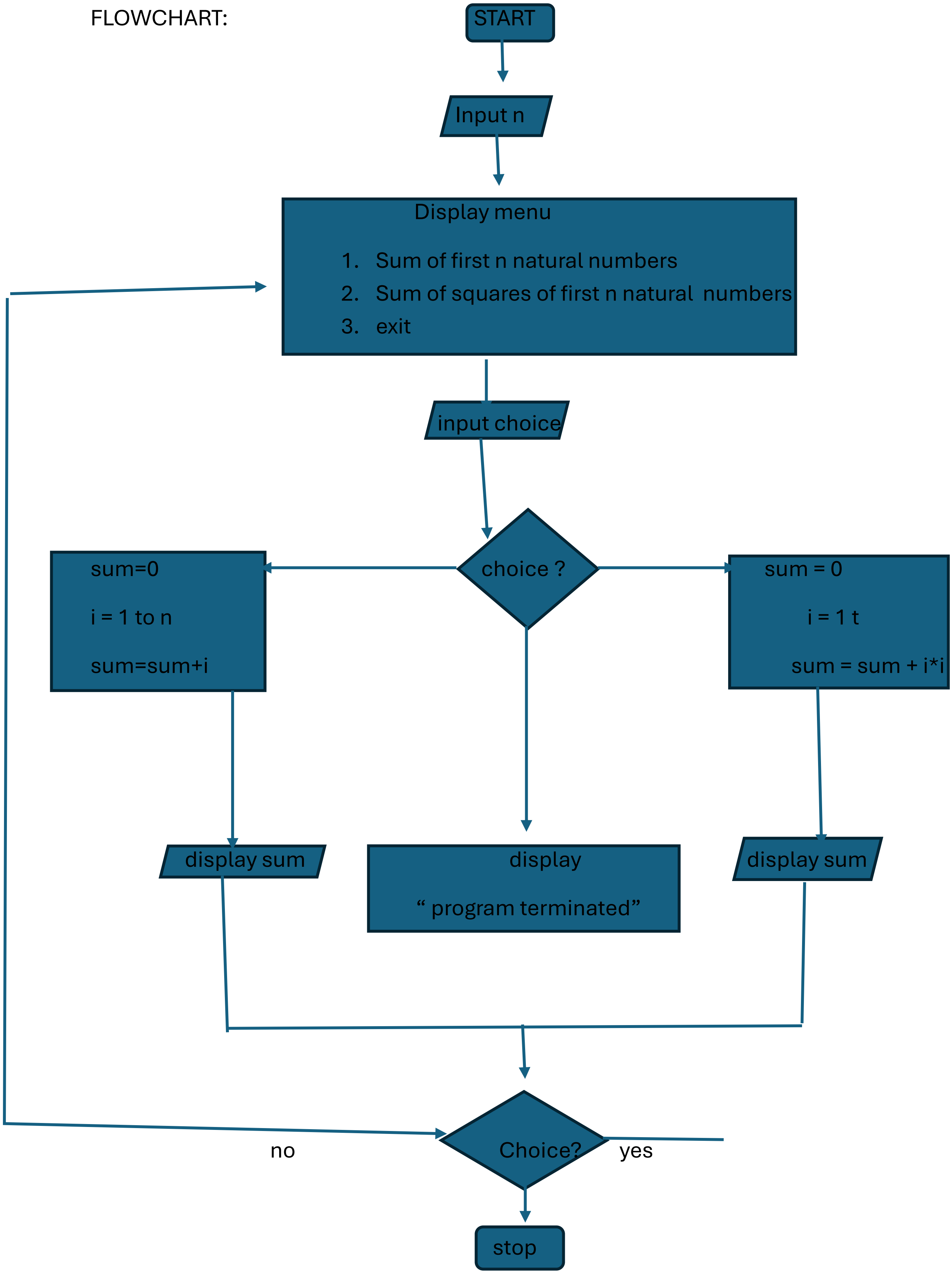
```
START

INPUT n

REPEAT
   DISPLAY "1. Sum of first n natural numbers"
   DISPLAY "2. Sum of squares"
   DISPLAY "3. Exit"

   INPUT choice

   IF choice = 1 THEN
```

HUNDRED DAYS OF CODE
DAY 9

```
    sum = 0

    FOR i = 1 TO n
      sum = sum + i
    END FOR

    DISPLAY sum

  ELSE IF choice = 2 THEN
    sum = 0

    FOR i = 1 TO n
      sum = sum + i * i
    END FOR

    DISPLAY sum

  ELSE IF choice = 3 THEN
    DISPLAY "Program terminated"

  ELSE
    DISPLAY "Invalid choice"
  END IF

UNTIL choice = 3

STOP
```

HUNDRED DAYS OF CODE
DAY 9

FLOWCHART:



HUNDRED DAYS OF CODE
DAY 9

C PROGRAM

```c
#include <stdio.h>


int main()

{

   int n, i, choice;

   long long sum;


   printf("Enter the value of n: ");

   scanf("%d", &n);


   if(n <= 0)

   {

      printf("Please enter a positive integer.\n");

      return 0;

   }


   do

   {

      printf("\n--- MENU ---\n");

      printf("1. Sum of first n natural numbers\n");

      printf("2. Sum of squares of first n natural numbers\n");

      printf("3. Exit\n");

      printf("Enter your choice: ");

      scanf("%d", &choice);


      switch(choice)
```

HUNDRED DAYS OF CODE
DAY 9

```
{
  case 1:
    sum = 0;

    for(i = 1; i <= n; i++)
      sum = sum + i;

    printf("Sum of first %d natural numbers = %lld\n",
        n, sum);
    break;

  case 2:
    sum = 0;

    for(i = 1; i <= n; i++)
      sum = sum + (long long)i * i;

    printf("Sum of squares of first %d natural numbers = %lld\n",
        n, sum);
    break;

  case 3:
    printf("Program terminated.\n");
    break;

  default:
    printf("Invalid choice! Please try again.\n");
```

```
    }

  } while(choice != 3);

  return 0;
}
```

TEST CASE TABLE

Test Case	n	Choice	Operation	Expected Output
TC-01	10	1	Natural number sum	55
TC-02	5	2	Sum of squares	55
TC-03	20	1	Natural number sum	210

COMPILATION AND EXECUTION

HUNDRED DAYS OF CODE
DAY 9

```
┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ ./natural
═════ MENU ═════
1. Sum of First n Natural Numbers
2. Sum of Squares of First n Natural Numbers
Enter your choice: 1
Enter the value of n: 10

Sum of first 10 natural numbers = 55
```

```
┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ ./natural
═════ MENU ═════
1. Sum of First n Natural Numbers
2. Sum of Squares of First n Natural Numbers
Enter your choice: 2
Enter the value of n: 5

Sum of squares of first 5 natural numbers = 55
```

HUNDRED DAYS OF CODE
DAY 9

```
┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ vi natural.c

┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ gcc natural.c -o natural

┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ ./natural
═════ MENU ═════
1. Sum of First n Natural Numbers
2. Sum of Squares of First n Natural Numbers
Enter your choice: 1
Enter the value of n: 10

Sum of first 10 natural numbers = 55

┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ ./natural
═════ MENU ═════
1. Sum of First n Natural Numbers
2. Sum of Squares of First n Natural Numbers
Enter your choice: 2
Enter the value of n: 5

Sum of squares of first 5 natural numbers = 55

┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ ./natural
═════ MENU ═════
1. Sum of First n Natural Numbers
2. Sum of Squares of First n Natural Numbers
Enter your choice: 1
Enter the value of n: 20

Sum of first 20 natural numbers = 210
```

HUNDRED DAYS OF CODE
DAY 9

QUESTION 3 : Write a **menu-driven** C program to calculate simple interest, compound amount, compound interest, and total payable amount for given principal, rate, and time.

Formulae

Simple Interest:

SI = (P × R × T) / 100

Total amount using Simple Interest:

Total Payable = P + SI

For annual compounding, the compound amount follows:

$$A = P\left(1 + \frac{r}{100}\right)^{t}$$

Compound Interest:

CI = A - P

where:

- P = Principal
- R = Annual rate of interest (%)
- T = Time in years
- A = Compound amount

Algorithm

1. Start.
2. Input principal P, rate R, and time T.
3. Display the menu:
    - Simple Interest
    - Compound Amount
    - Compound Interest
    - Total Payable Amount
    - Exit
4. Read the user's choice.

5. Perform the selected calculation.
6. Display the result.
7. Repeat until Exit is selected.
8. Stop.

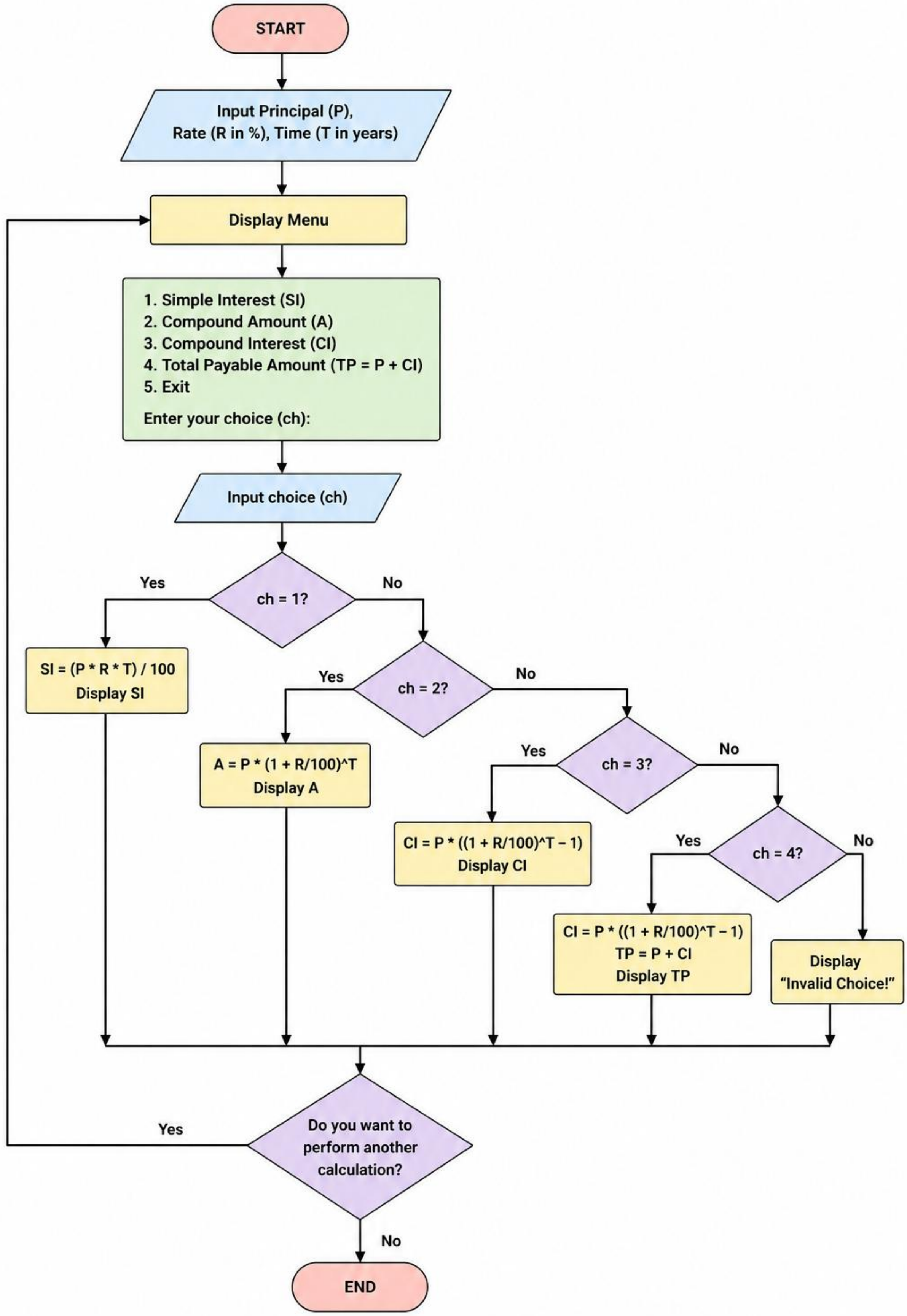
PSEUDOCODE

```
START

INPUT P, R, T

REPEAT

   DISPLAY "1. Simple Interest"
   DISPLAY "2. Compound Amount"
   DISPLAY "3. Compound Interest"
   DISPLAY "4. Total Payable Amount"
   DISPLAY "5. Exit"

   INPUT choice

   SWITCH choice

      CASE 1:
         SI = (P * R * T) / 100
         DISPLAY SI

      CASE 2:
         A = P * (1 + R/100)^T
```

```
    DISPLAY A

  CASE 3:
    A = P * (1 + R/100)^T
    CI = A - P
    DISPLAY CI

  CASE 4:
    SI = (P * R * T) / 100
    Total = P + SI
    DISPLAY Total

  CASE 5:
    DISPLAY "Program terminated"

  DEFAULT:
    DISPLAY "Invalid choice"

UNTIL choice = 5

STOP
```

FLOWCHART

HUNDRED DAYS OF CODE
DAY 9



HUNDRED DAYS OF CODE
DAY 9

REQUIRED C PROGRAM :

```
 #include <stdio.h>

#include <math.h>


int main(void)

{

   int choice, time;

   double principal, rate;

   double simpleInterest, amount, compoundInterest;


   do

   {

      printf("\n===== INTEREST CALCULATION MENU =====\n");

      printf("1. Simple Interest\n");

      printf("2. Compound Amount\n");

      printf("3. Compound Interest\n");

      printf("4. Total Payable Amount\n");

      printf("5. Exit\n");

      printf("Enter choice: ");

      scanf("%d", &choice);


      if (choice >= 1 && choice <= 4)

      {

         printf("Enter principal: ");

         scanf("%lf", &principal);


         printf("Enter annual rate (%%): ");
```

HUNDRED DAYS OF CODE
DAY 9

```
scanf("%lf", &rate);


printf("Enter time (years): ");

scanf("%d", &time);


if (principal < 0 || rate < 0 || time < 0)

{

   printf("Principal, rate and time must be non-negative.\n");

}

else

{

   simpleInterest =

      principal * rate * time / 100.0;


   amount =

      principal * pow(1.0 + rate / 100.0, time);


   compoundInterest = amount - principal;


   switch (choice)

   {

      case 1:

         printf("Simple Interest = %.2f\n",

              simpleInterest);

         break;


      case 2:
```

```
                printf("Compound Amount = %.2f\n",
                    amount);
                break;

              case 3:
                printf("Compound Interest = %.2f\n",
                    compoundInterest);
                break;

              case 4:
                printf("Total Payable Amount = %.2f\n",
                    principal + simpleInterest);
                break;
            }
          }
        }
        else if (choice != 5)
        {
          printf("Invalid choice!\n");
        }

    } while (choice != 5);

    printf("Program terminated.\n");

    return 0;
}
```

HUNDRED DAYS OF CODE
DAY 9

TEST CASE TABLE :

TEST CASE	INPUT(P, R, T, CHOICE)	EXPECTED OUTPUT	ACTUAL OUTPUT	RESULT
1	10000, 10,2,1	SI = = ₹2000.00	SAME	PASS
2	10000, 10, 2, 1	AMOUNT = = ₹2000.00	SAME	PASS
3	10000,10, 2, 3	CI = ₹2100.0	SAME	PASS
4	1000, 10, 2, 4	TOTAL PAYABLE = ₹12000.00	SAME	PASS

COMPILATION AND EXECUTION

HUNDRED DAYS OF CODE
DAY 9

```
┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ ./interest
═════ MENU ═════
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
Enter your choice: 2
Enter Principal Amount: 10000
Enter Rate of Interest: 10
Enter Time (Years): 2

Compound Amount = Rs. 12100.00
```

```
┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ ./interest
═════ MENU ═════
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
Enter your choice: 3
Enter Principal Amount: 10000
Enter Rate of Interest: 10
Enter Time (Years): 2

Compound Interest = Rs. 2100.00
```

```
┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ ./interest
═════ MENU ═════
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
Enter your choice: 1
Enter Principal Amount: 10000
Enter Rate of Interest: 10
Enter Time (Years): 2

Simple Interest = Rs. 2000.00
```

HUNDRED DAYS OF CODE
DAY 9

```
┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ vi interest.c

┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ gcc interest.c -o interest -lm

┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ ./interest
═══ MENU ═══
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
Enter your choice: 1
Enter Principal Amount: 10000
Enter Rate of Interest: 10
Enter Time (Years): 2

Simple Interest = Rs. 2000.00

┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ ./interest
═══ MENU ═══
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
Enter your choice: 2
Enter Principal Amount: 10000
Enter Rate of Interest: 10
Enter Time (Years): 2

Compound Amount = Rs. 12100.00
```

```
┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ ./interest
═══ MENU ═══
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
Enter your choice: 4
Enter Principal Amount: 10000
Enter Rate of Interest: 10
Enter Time (Years): 2

Total Payable Amount = Rs. 12000.00
```

```
┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ ./interest
═══ MENU ═══
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
Enter your choice: 3
Enter Principal Amount: 10000
Enter Rate of Interest: 10
Enter Time (Years): 2

Compound Interest = Rs. 2100.00

┌──(kali㉿kali)-[~/Desktop/exp8]
└─$ ./interest
═══ MENU ═══
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
Enter your choice: 4
Enter Principal Amount: 10000
Enter Rate of Interest: 10
Enter Time (Years): 2

Total Payable Amount = Rs. 12000.00
```